Algorithms and Optimization Project Report

Group 23
-João Patrício, 20231645
-Rodrigo Silva, 20231602
-Vasco Rodrigues, 20231676

Introduction

The goal of this project is to optimize the routes of flying robots in order to reduce the fuel spent or the traveled distance. The robots' flight is constrained by a number of factors, either physical or operational. This report shows how we formalized the problem, solved it and interpreted its solution results.

Problem formalization

To solve this optimization problem we need that our cost functions and constraint sets are convex. This section of the report shows how we formalized the problem and showed that its easily solvable. All the boxed questions were solved in this section, by the order they appeared in the guidelines.

Cost function: fuel consumption

Convexity of the Cost Function

Question. Is the cost function

$$\sum_{k=0}^{n} f\left(\|\mathbf{u}(k)\|\right)$$

convex, given that f is a monotonically increasing convex function? Justify.

The fuel consumption cost function, F(n), is a sum of composite functions.
A composite function gof is convex, if g is convex and non decreasing and f is convex.
The proof is:
Let f be a convex function and g non decreasing:

$gof(\lambda x_1 + (1 - \lambda)x_2) =$
$g(f(\lambda x_1 + (1 - \lambda)x_2)) \leq$
$g(\lambda f(x_1) + (1 - \lambda)f(x_2)) \leq$ (since f is convex and non decreasing)

$\lambda g(f(x_1)) + (1 - \lambda)g(f(x_2)) =$ (since g is convex)
$\lambda gof(x_1) + (1 - \lambda)gof(x_2)$

$gof(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda gof(x_1) + (1 - \lambda)gof(x_2)$ is proved, gof is convex.

In our cost function, we have f*o*g where f is stated to be non decreasing and convex and g is the euclidean norm of a vector, which is convex (as every other norm).
This is the proof:
Let f(x) = ||x||. Then:

$f(\lambda x_1 + (1 - \lambda)x_2) =$
$||\lambda x_1 + (1 - \lambda)\, x_2|| \leq$
$||\lambda x_1|| + ||(1 - \lambda)\, x_2|| =$ (Triangular Inequality)
$\lambda||x_1|| + (1 - \lambda)||x_2|| =$ (for λ in [0, 1])
$\lambda f(x_1) + (1 - \lambda)f(x_2)$

$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$ is proved, f(x)=||x|| is convex.

Now that we know that the cost function is actually a sum of convex functions, the cost function will be convex since the sum of convex functions is also convex. The proof of this is very simple: Let h(x) = f(x) + g(x) where f and g are convex functions. Then:

$h(\lambda x_1 + (1 - \lambda)x_2) =$
$f(\lambda x_1 + (1 - \lambda)\, x_2) + g(\lambda x_1 + (1 - \lambda)\, x_2) \leq$
$\lambda f(x_1) + (1 - \lambda)f(x_2) + \lambda g(x_1) + (1 - \lambda)g(x_2) =$ (since f and g are convex)
$\lambda h(x_1) + (1 - \lambda)h(x_2)$

for λ in [0, 1]

Since the fuel consumption cost function, F(n), is convex, we know that there are no local minimums and the optimization algorithms will find the global minimum.

Cost function: distance traveled

Convexity of Distance-Based Cost

Question. Is the cost function

$$\sum_{k=1}^{n} \|\mathbf{x}(k) - \mathbf{x}(k-1)\|$$

convex? What about the squared version? What about the L_1 norm version?

This time the cost function D(n), now related to distance traveled, is a sum of functions, so to be convex its elements must also be convex. Its elements are euclidean norm functions, which are convex.

If its elements are the norm squared, then it will still be convex because the norm squared is a convex function. Here is the proof: Let f(x) = $||x||^2$. Then:

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2) \Leftrightarrow$$
$$||\lambda x_1 + (1-\lambda)\, x_2||^2 \leq \lambda ||x_1||^2 + (1-\lambda)||x_2||^2 \Leftrightarrow$$
$$||\lambda x_1||^2 + 2\lambda(1-\lambda)x_1^T x_2 + ||(1-\lambda)x_2||^2 \leq \lambda||x_1||^2 + (1-\lambda)||x_2||^2$$
$$0 \leq \lambda||x_1||^2 + (1-\lambda)||x_2||^2 - \lambda^2||x_1||^2 - 2\lambda(1-\lambda)x_1^T x_2 - (1-\lambda)^2||x_2||^2 \Leftrightarrow$$
$$0 \leq ||x_1||^2(\lambda - \lambda^2) + ||x_2||^2(1 - \lambda - (1-\lambda)^2) - 2(\lambda - \lambda^2)x_1^T x_2 \Leftrightarrow$$
$$0 \leq ||x_1||^2(\lambda - \lambda^2) + ||x_2||^2(1 - \lambda - 1 + 2\lambda - \lambda^2) - 2(\lambda - \lambda^2)x_1^T x_2 \Leftrightarrow$$
$$0 \leq ||x_1||^2(\lambda - \lambda^2) + ||x_2||^2(\lambda - \lambda^2) - 2(\lambda - \lambda^2)x_1^T x_2 \Leftrightarrow$$
$$0 \leq (\lambda - \lambda^2)(||x_1||^2 + ||x_2||^2 - 2x_1^T x_2) \Leftrightarrow$$
$$0 \leq (\lambda - \lambda^2)(||x_1| + ||x_2||)^2$$

for λ in [0, 1]

The L1 norm function won't change a thing about convexity, for all norms are convex.

Now we have proved that the distance traveled cost function D(n) is convex, which once again will let us find its global minimum, for there are no local ones.

Note about the cost functions

Using both of the cost functions, it's possible to create one new cost function that can be used to find solutions that try to minimize both fuel consumption and the distance traveled. This new cost function can be represented as:

$$C(n) = F(n) + D(n)$$

This cost function, being the sum of two convex functions, will be convex, so it's possible to easily find its global minimum.

However, it's very likely that the solution that best minimizes fuel consumption won't be the one that best minimizes distance traveled. For whatever reason, minimizing one of these things might be much more important than the other. If we assign a weight for each of the parts of C, we can choose how much one thing should be given for the other. Our function then can be represented as:

$C(n, \Delta) = \Delta F(n) + (1 - \Delta)D(n)$

For Δ in [0,1]

Obviously, if Δ is 1 or 0 our new cost function will only try to minimize fuel or distance respectively. We will experiment different values of Δ to see how the best path changes according to the chosen priorities.

Physical constraint: Newton's law

Discretization Task

Question 1. Discretize the continuous-time dynamics in equation (14) using a time step of size h, following the approach presented in class.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?
The optimization variables involved in the formulation of this constraint are:

- $\mathbf{x}_i(k) \in \mathbb{R}^3$: position of robot i at time step k
- $\mathbf{v}_i(k) \in \mathbb{R}^3$: velocity of robot i at time step k
- $\mathbf{u}_i(k) \in \mathbb{R}^3$: thrust vector (control input) applied to robot i at time step k.

The robot 's dynamics must obey the Newton' s law. This law act as a constraint, that can be defined as:

$$m_i \frac{dv_i}{d\tau}(t) = u_i(t) - \beta v_i(t) + m_i g \approx$$
$$m_i \frac{v_i((k+1)h) - v_i(kh)}{h} = u_i(kh) - \beta v_i(kh) + m_i g \Leftrightarrow$$
$$v_i((k+1)h) - v_i(kh) = \frac{h}{m_i}(u_i(kh) - \beta v_i(kh) + m_i g) \Leftrightarrow$$
$$v_i((k+1)h) = \frac{h}{m_i}(u_i(kh) - \beta v_i(kh) + m_i g) + v_i(kh)$$

$$\frac{dx_i}{d\tau}(t) = v_i(t) \approx$$

$$\frac{x_i((k+1)h) - x_i(kh)}{h} = v_i(kh) \Leftrightarrow$$
$$x_i((k+1)h) - x_i(kh) = h(v_i(kh)) \Leftrightarrow$$
$$x_i((k+1)h) = h(v_i(kh)) + x_i(kh)$$

This sets are linear which makes them convex, as proven here:

$S = \{x : Ax = b\}$

$x_1, x_2 \in S$ i.e

$Ax_1 = b$ and $Ax_2 = b$

then:

$$A(\lambda x_1 + (1 - \lambda) x_2) =$$
$$A\lambda x_1 + A(1 - \lambda) x_2 =$$
$$\lambda b + (1 - \lambda) b = b$$

for λ in [0, 1]

Since both of these are convex, the problem can still be easily minimized.

Operation constraint: wireless communication

Communication Constraint

Question 1. Formalize this constraint. The main optimization variable involved is the discrete-time position of the robots, denoted by $\mathbf{x}_i(k) \in \mathbb{R}^3$ across time $k = 0, 1, 2, \ldots, n$.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?

The constraint can be represented by the following set:

$S = \{x : ||x_i(k) - x_j(k)|| \leq d, \forall i, j : i \neq j, \forall k\}$

This set is convex, and here is the proof:

$S = \{x : ||x - y|| \leq d\}$

$(x_1, y_1), (x_2, y_2) \in S$ i.e

$||x_1 - y_1|| \leq d$ and $||x_2 - y_2|| \leq d$

Then:

$$||\lambda x_1 + (1 - \lambda)\, x_2 - (\lambda y_1 + (1 - \lambda)\, y_2)|| =$$
$$||\lambda(x_1 - y_1) + (1 - \lambda)(x_2 - y_2)|| \leq$$
$$\lambda||x_1 - y_1|| + (1 - \lambda)||x_2 - y_2|| \leq d$$

for λ in [0, 1]

Being this set convex, this constraint won't harm the convexity of our cost functions.

Waypoint track

Waypoint Constraint

Question 1. Formally write the set of constraints enforcing that robot i must pass through each of its assigned waypoints at the corresponding times.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?

This constraint states that there are waypoints assign to each robot that must be reached at a certain w time. That can be written as the following set:

$$S = \{x : x_i(k_{il}) = w_{il}, \forall\, i, l\}$$

This set is linear, and we have already proved that these types of sets are convex, which means this set will not impact the problem's solvability.

Operation constraint: maximum altitude

Altitude Constraint

Question 1. Express the altitude constraint mathematically. The relevant optimization variable is the robot's position $\mathbf{x}_i(k) = [x_i(k), y_i(k), z_i(k)]^\top \in \mathbb{R}^3$. You may assume the vertical axis corresponds to the z-coordinate.
Question 2. Is this constraint convex? That is, does the set of points satisfying this constraint form a convex subset of the optimization space?

This constraint states that the coordinate z of the position vector x of the robot must not surpass a certain height h, and can be represented as the convex set:

$S = \{z : z_i(k) \leq h, \forall i, k\}$

It is very simple to prove its convexity:

$S = \{x : x \leq h\}$

$x_1, x_2 \in S$ i.e

$x_1 \leq h$ and $x_2 \leq h$

Then:

$\lambda x_1 + (1 - \lambda) x_2 \leq h$

for λ in [0, 1]

Once again, our problem can still be easily solved because this constraint is a convex set.

Note about the constraints

Because all the constraints in this problem are convex sets, not only can they be applied but they can be applied at the same time. This is because the intersection of a convex set is a convex set. The proof of this statement goes as follow:

$S = S_1 \cap S_2$

$x_1, x_2 \in S_1$ and $x_1, x_2 \in S_2$

Therefore:

$\lambda x_1 + (1 - \lambda) x_2 \in S_1$ and $\lambda x_1 + (1 - \lambda) x_2 \in S_2$

Obviously this holds for any finite n sets.

Formalization

Joining everything, we can write the whole problem as:

Minimize: $C(n, \Delta) = \Delta F(n) + (1 - \Delta)D(n)$

Subject to: $v_i((k + 1)h) = \frac{h}{m_i}(u_i(kh) - \beta v_i(kh) + m_i g) + v_i(kh)$,

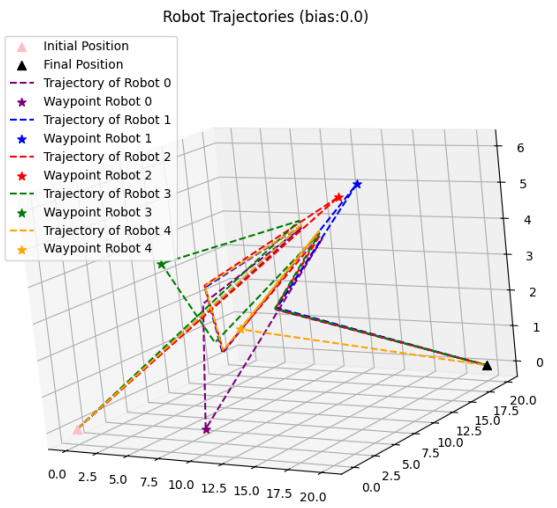
$x_i((k + 1)h) = h(v_i(kh)) + x_i(kh)$,

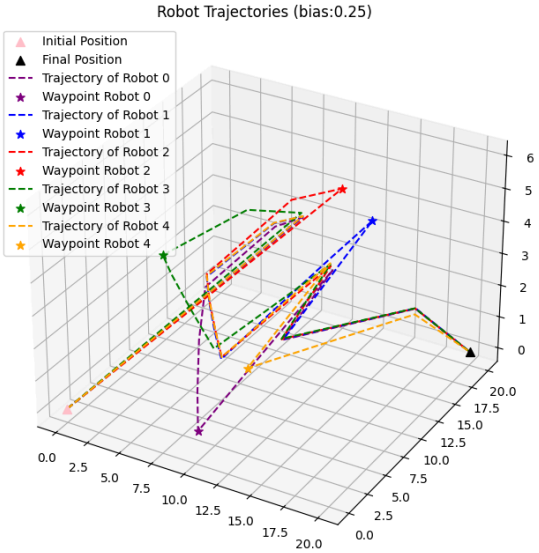
$$||x_i(k) - x_j(k)|| \leq d,$$
$$x_i(k_{il}) = w_{il},$$
$$z_i(k) \leq h$$

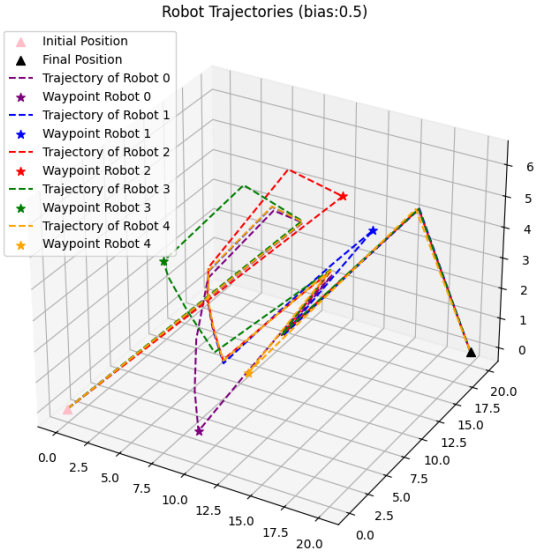
Results

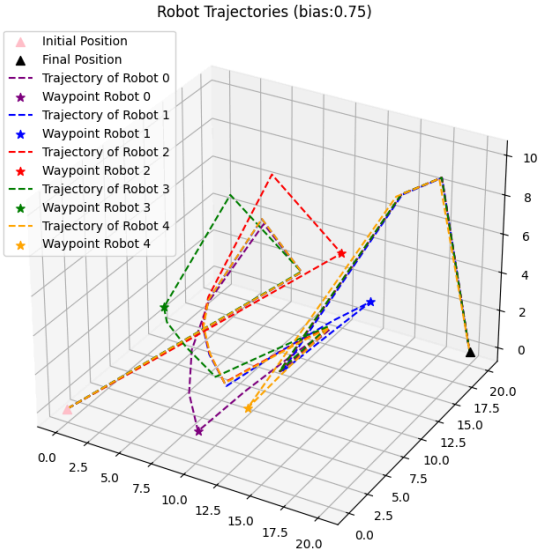
Now that we have used cvxpy to get the optimal trajectories, let's analyse what it means in the problem's context, using a variety of different plots. We will compare the result using the following bias values: 0, 0.25, 0.5, 0.75 and 1. As a reminder, the higher the bias the more important the fuel cost function will be, and the lower the more important the distances will be.

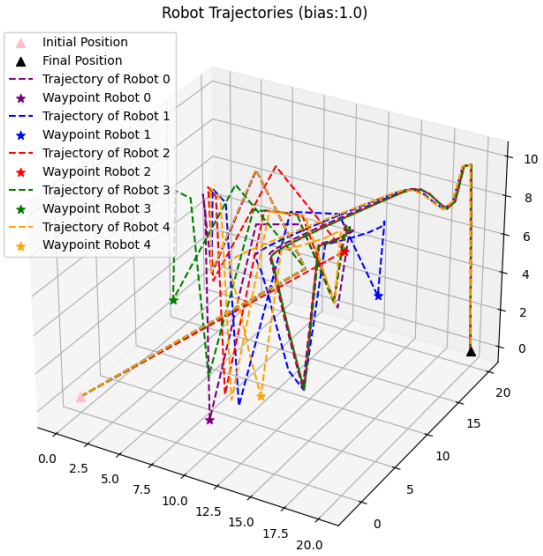
Trajectory plots





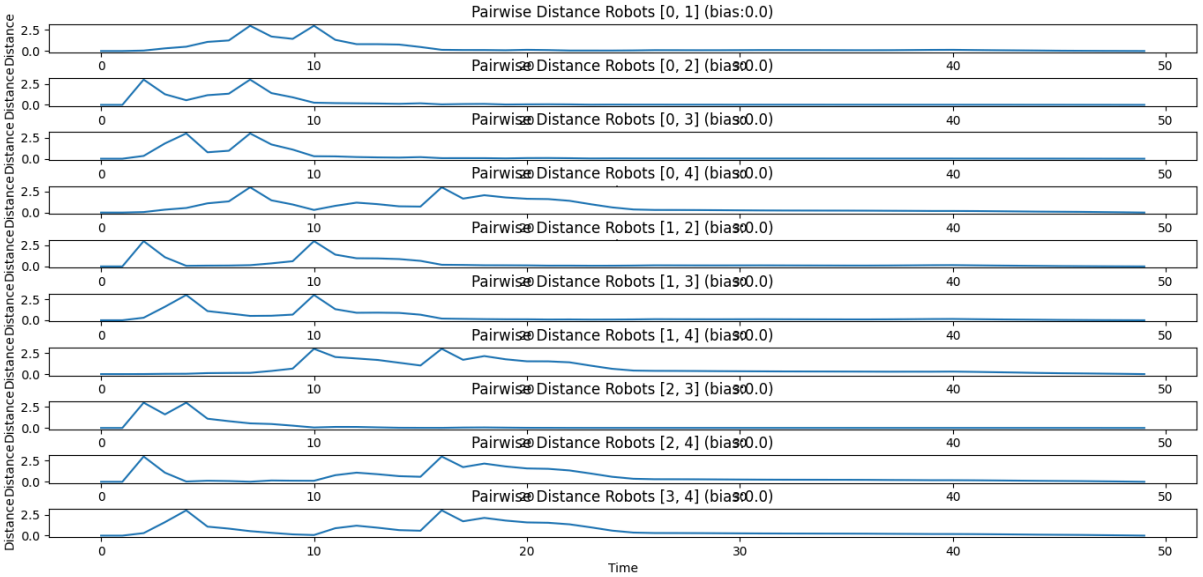


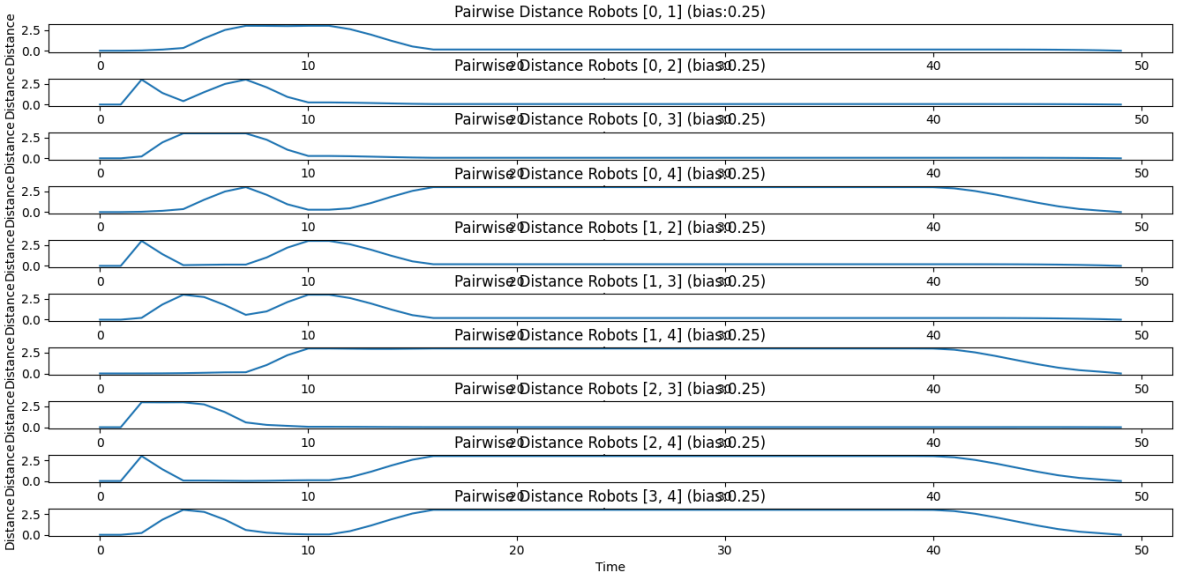


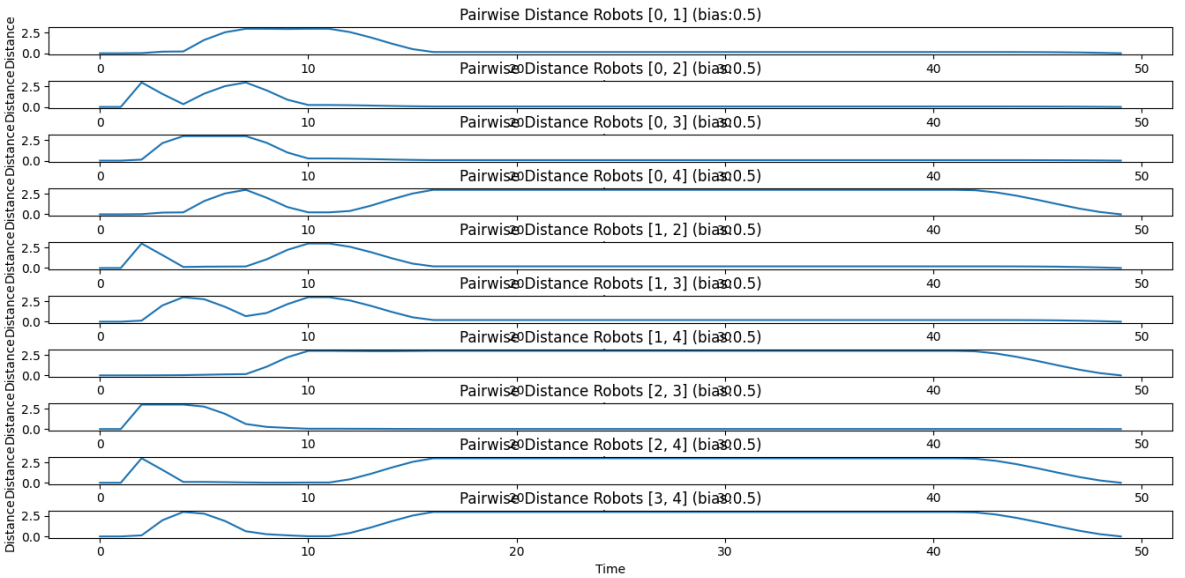


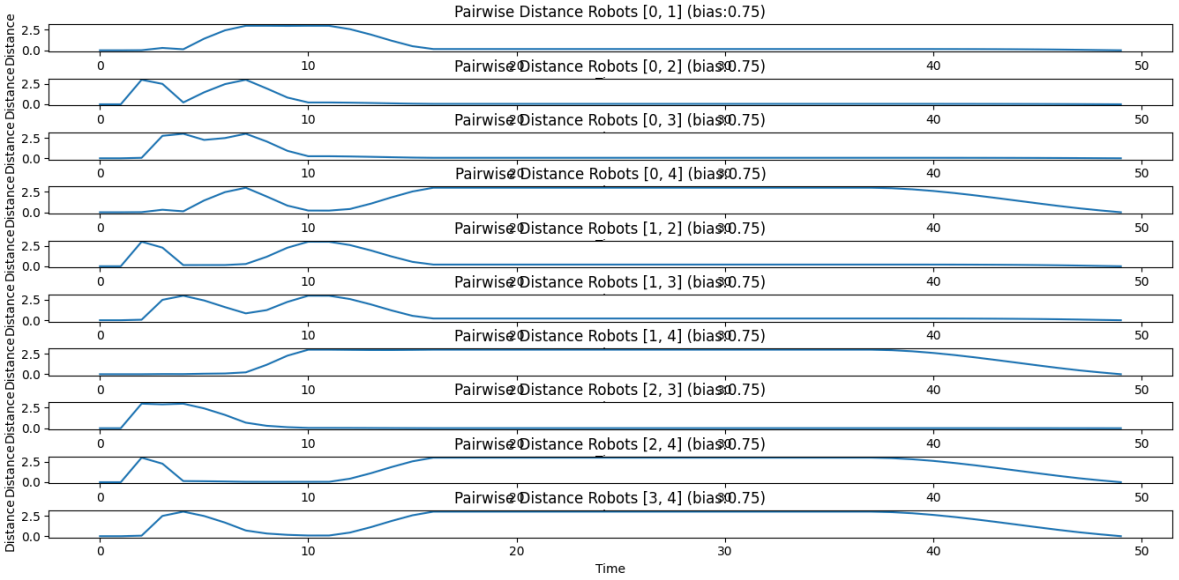
We can see that a lower bias will make the robots choose the straightest path possible to get to the waypoints, utilizing high magnitude and fuel consuming thrusts, therefore minimizing distance. On the other hand, when the bias is high the robots will take “v” shaped paths between its waypoints, using its momentum and letting themselves descend to spare fuel only to rise again when needed.

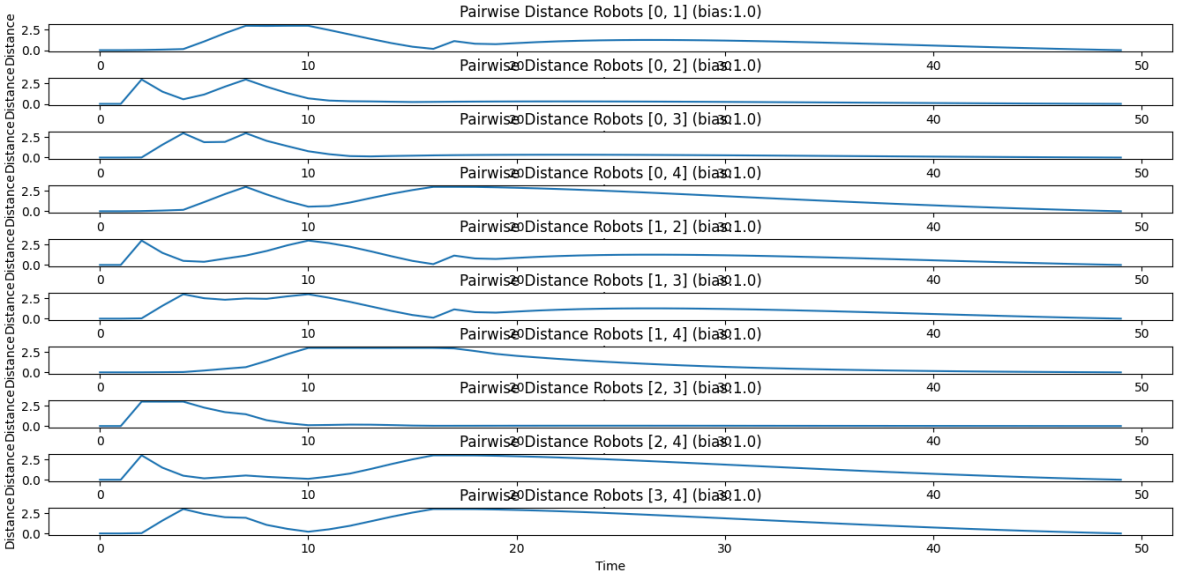
Pairwise Distance Plots





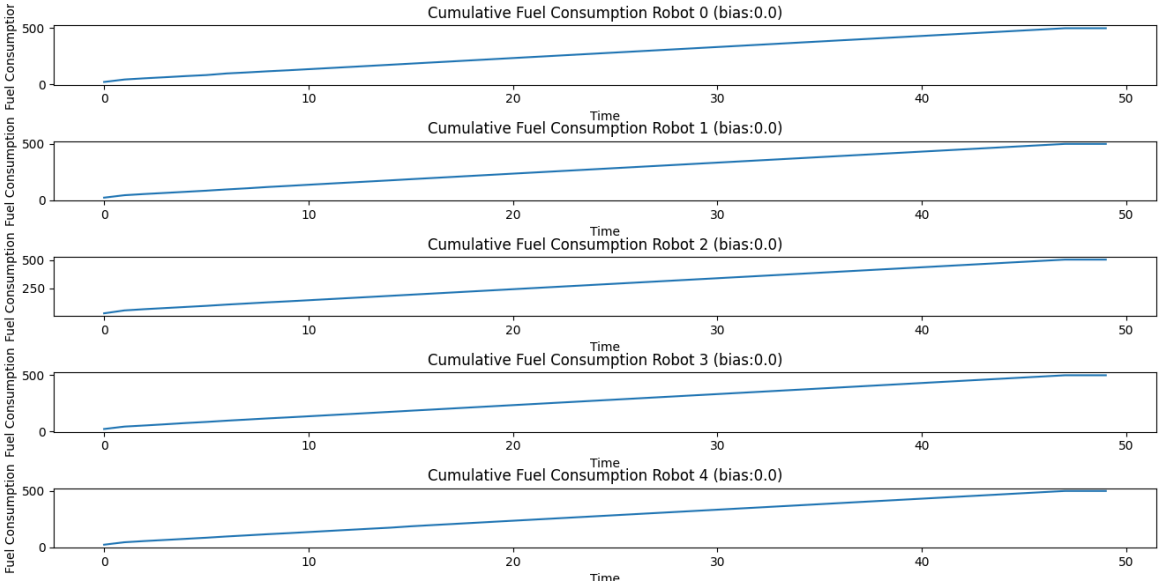


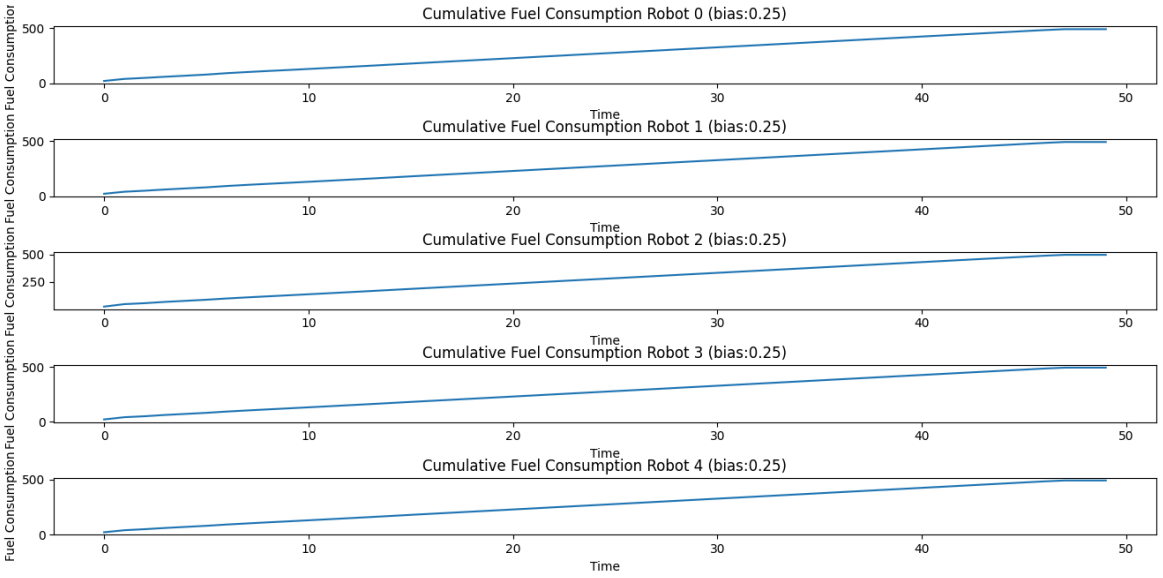


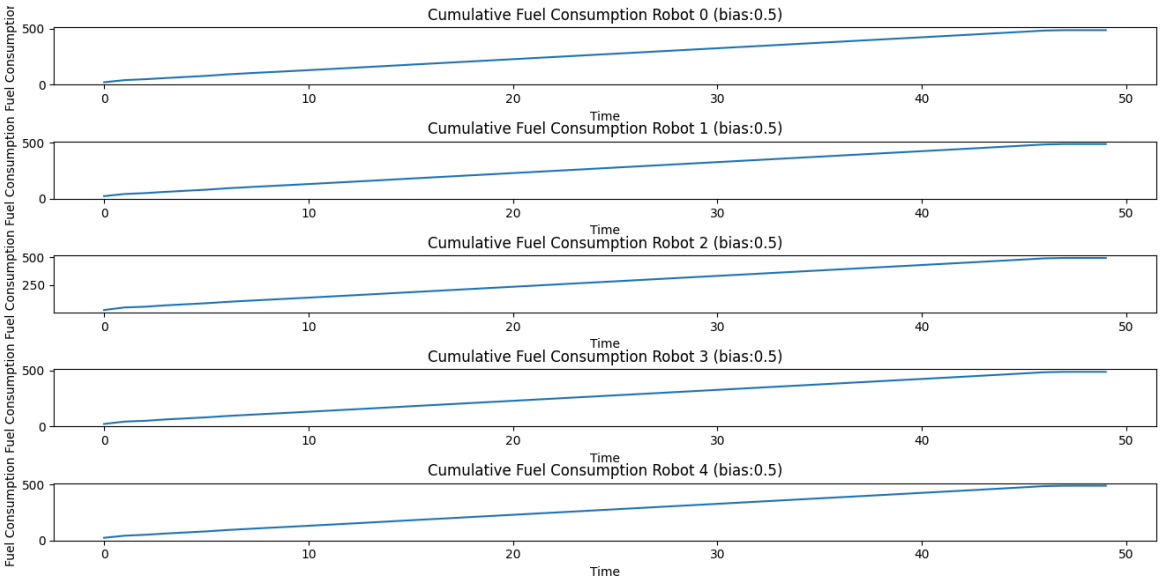


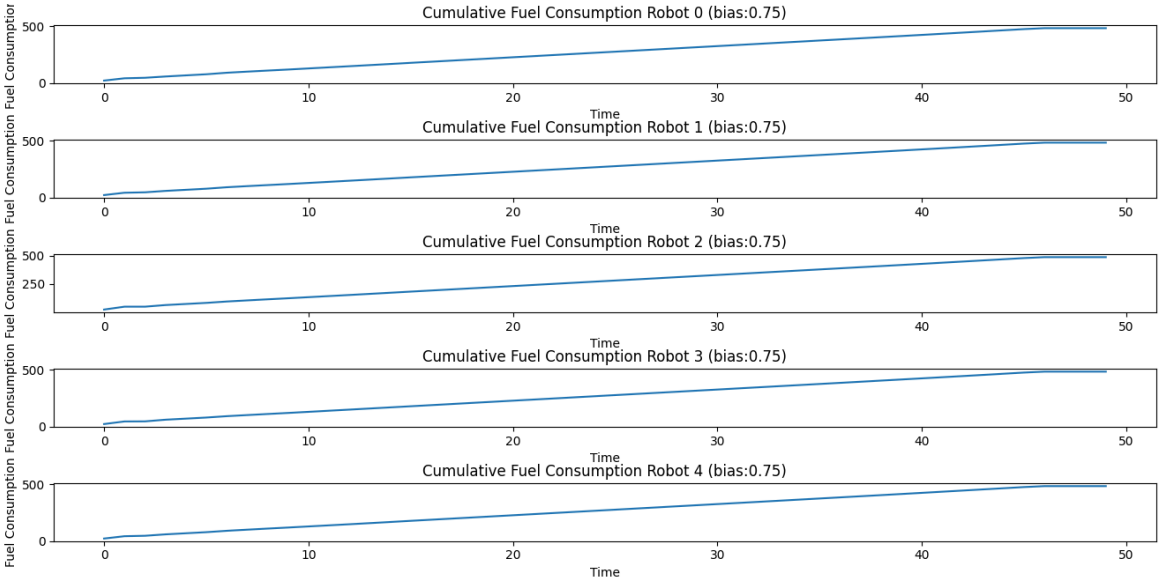
At low bias, robots prioritize reaching their waypoints quickly, causing them to spread out initially. Then they come together as they approach the final destination, maintaining communication. When bias increases, robots start focusing more and more on fuel efficiency, which often results in them moving more carefully and potentially alternating their relative positions or spreading less drastically to conserve fuel.

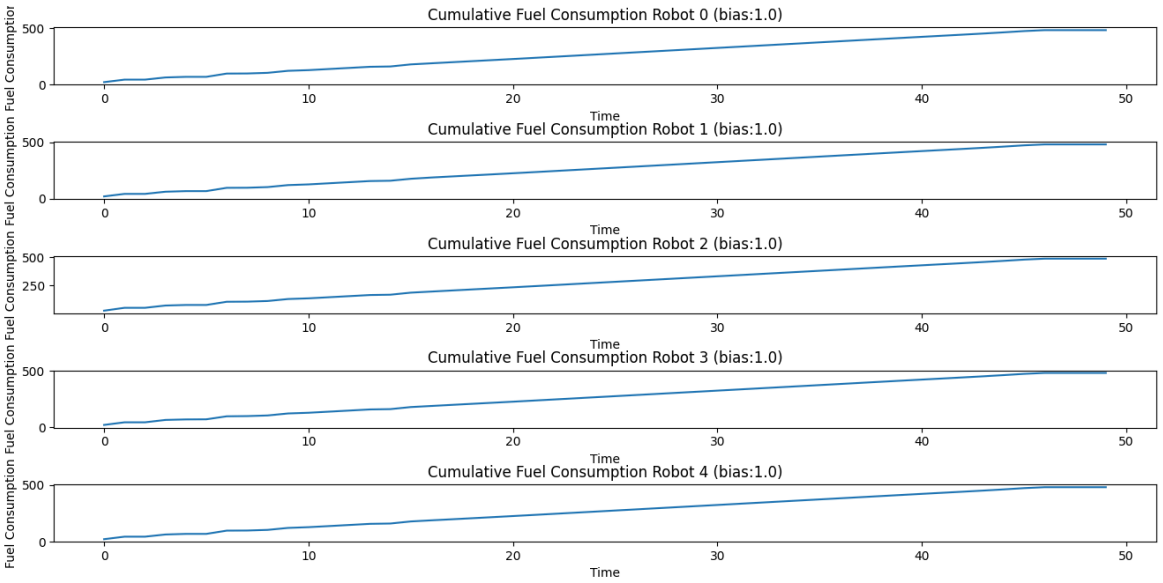
Cumulative Fuel Consumption Plots





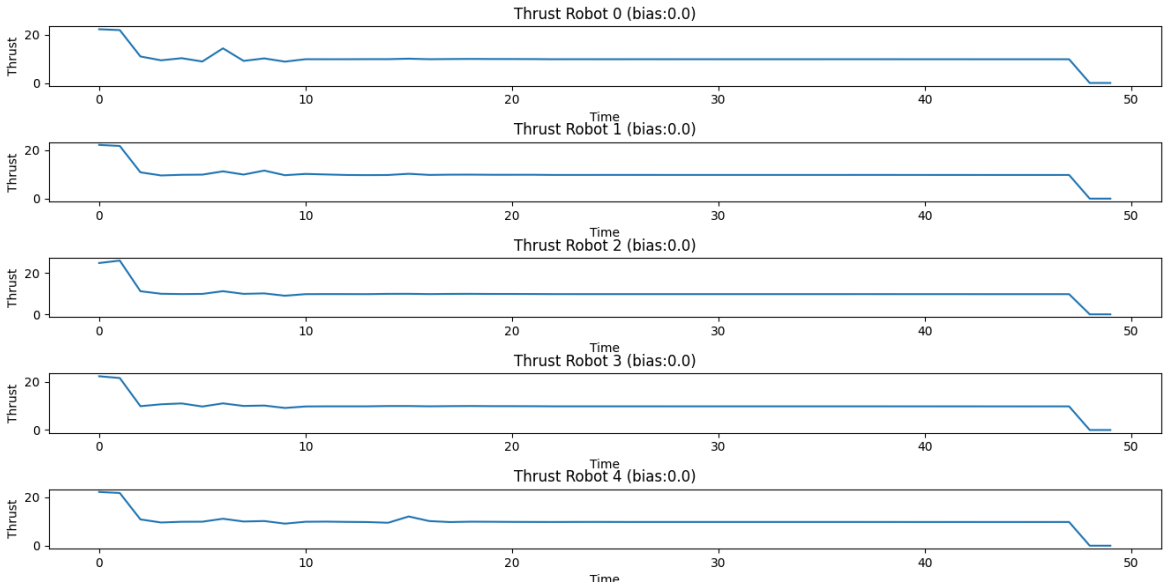

Cumulative Fuel Consumption Robot 0 (bias:0.5)
Cumulative Fuel Consumption Robot 1 (bias:0.5)
Cumulative Fuel Consumption Robot 2 (bias:0.5)
Cumulative Fuel Consumption Robot 3 (bias:0.5)
Cumulative Fuel Consumption Robot 4 (bias:0.5)
Fuel Consumption
Time

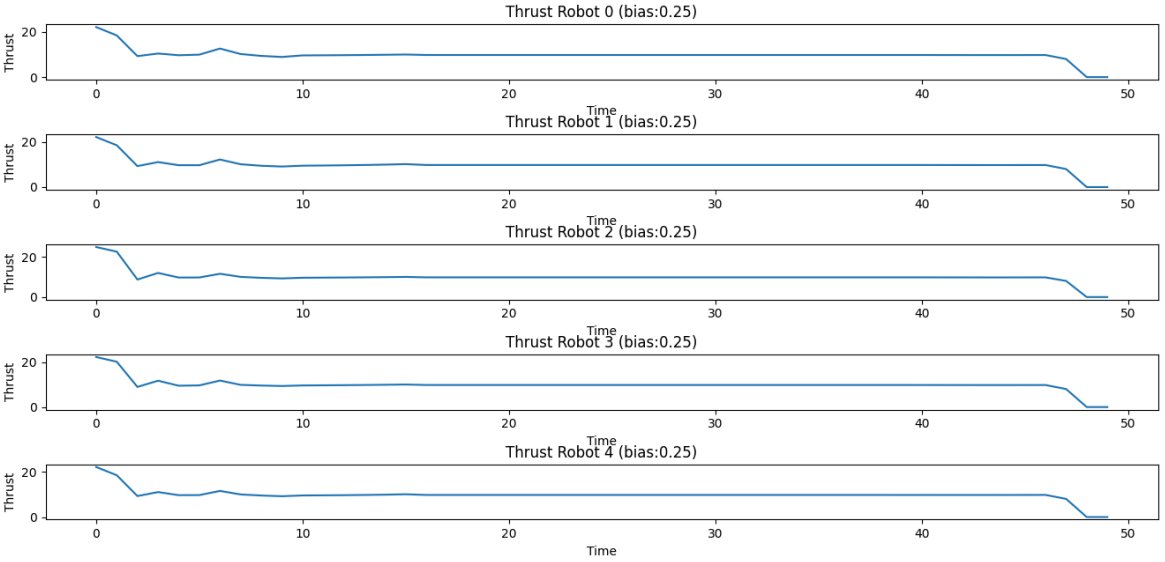

Cumulative Fuel Consumption Robot 0 (bias:0.75)
Cumulative Fuel Consumption Robot 1 (bias:0.75)
Cumulative Fuel Consumption Robot 2 (bias:0.75)
Cumulative Fuel Consumption Robot 3 (bias:0.75)
Cumulative Fuel Consumption Robot 4 (bias:0.75)
Fuel Consumption
Time

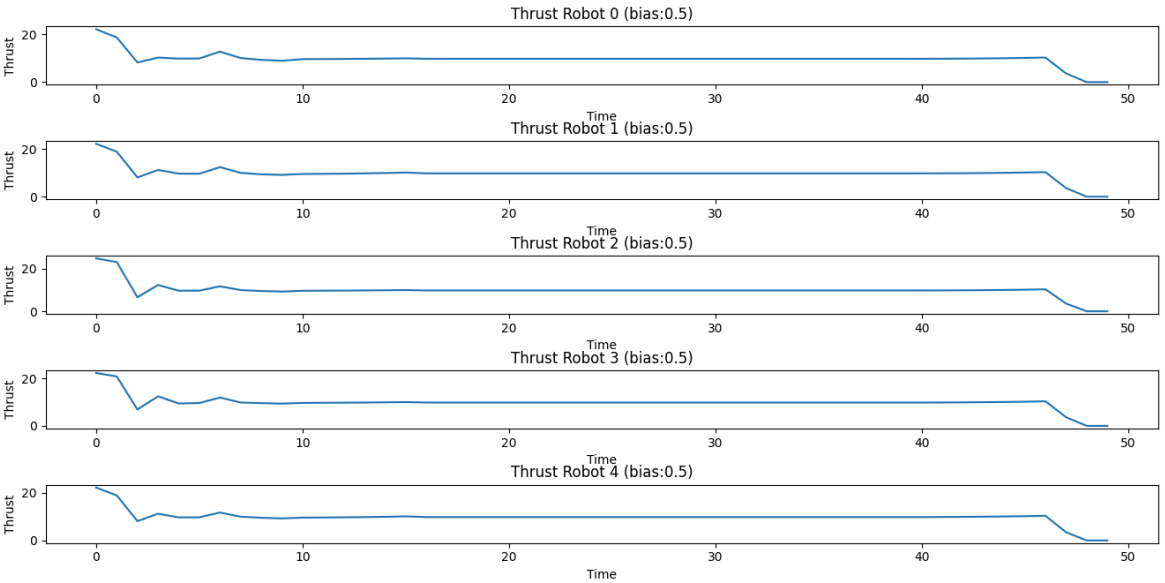

Cumulative Fuel Consumption Robot 0 (bias:1.0)
Cumulative Fuel Consumption Robot 1 (bias:1.0)
Cumulative Fuel Consumption Robot 2 (bias:1.0)
Cumulative Fuel Consumption Robot 3 (bias:1.0)
Cumulative Fuel Consumption Robot 4 (bias:1.0)
Fuel Consumption
Time

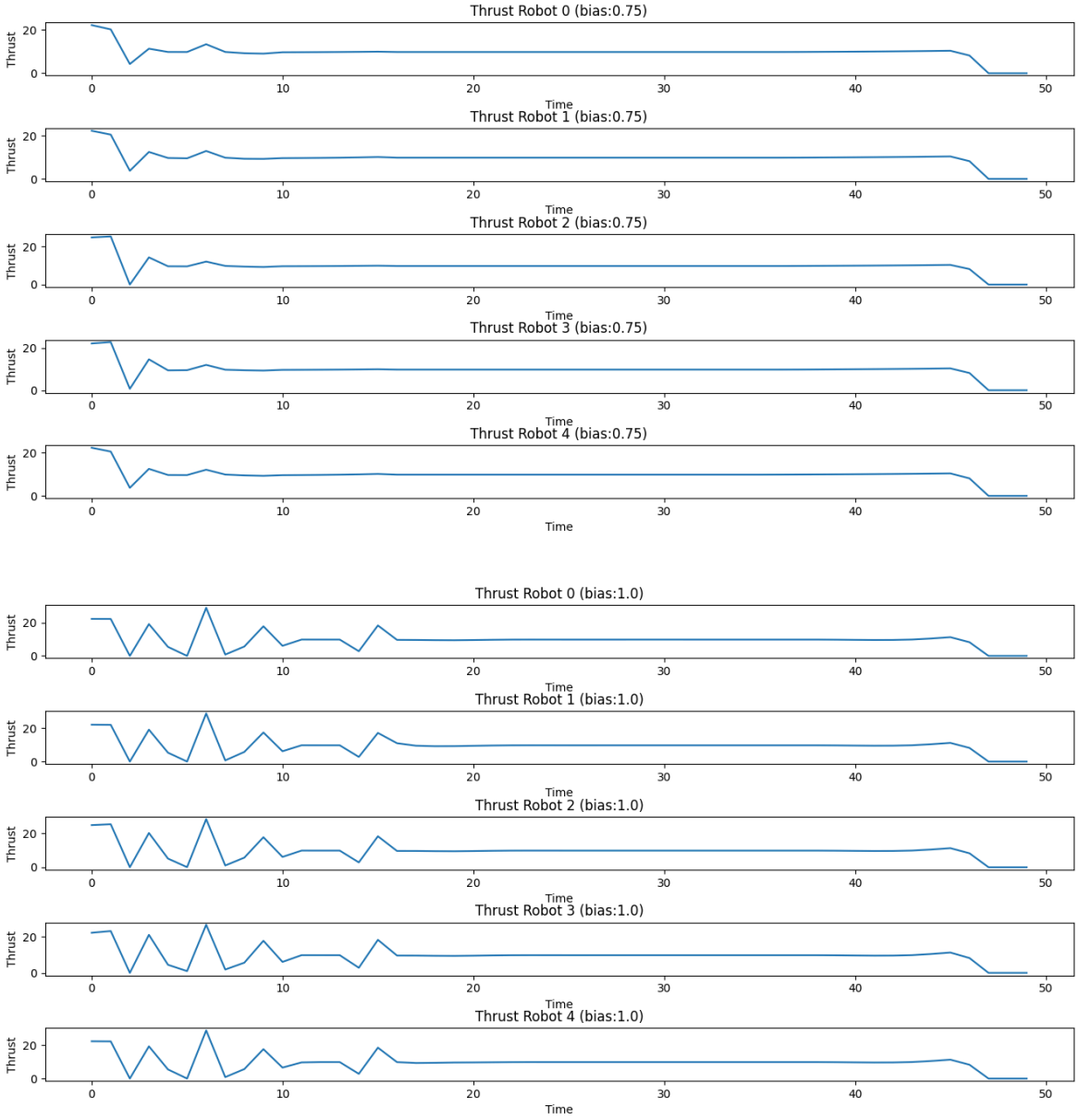
When the bias is low, the cumulative fuel consumption tends to be higher. This is evident in the steeper slopes and higher final values on the graphs for bias=0.0 and bias=0.25 compared to higher bias values. Robots are willing to use more fuel to achieve more direct or shorter paths. As the bias increases towards 1.0, the cumulative fuel consumption for each robot is lower. The slopes of these graphs are less steep, and the final fuel consumption values are somewhat reduced, even though one robot may be sacrificed to keep the total fuel consumption lower.

Thrust Plots




Thrust Robot 0 (bias:0.25)
Thrust Robot 1 (bias:0.25)
Thrust Robot 2 (bias:0.25)
Thrust Robot 3 (bias:0.25)
Thrust Robot 4 (bias:0.25)
Thrust
Time
0
10
20
30
40
50


Thrust Robot 0 (bias:0.5)
Thrust Robot 1 (bias:0.5)
Thrust Robot 2 (bias:0.5)
Thrust Robot 3 (bias:0.5)
Thrust Robot 4 (bias:0.5)
Thrust
Time
0
10
20
30
40
50



Robots, at lower bias, apply a consistently higher level of thrust throughout the mission, with general fluctuations, to achieve faster movements and minimize overall travel distance. When bias starts getting higher, robots exhibit an initial, pronounced spike in thrust, likely for waypoint alignment. However, the thrust then drops to a significantly lower and more stable baseline, aiming to conserve fuel for the remainder of the trajectory.

Plots Summary

This simulation demonstrates a trade-off between travel distance and conserving fuel, managed by a “bias” parameter. At low bias (distance prioritizing), robots show higher, consistent thrusts leading to greater cumulative fuel consumption. They initially spread out more to reach waypoints quickly, then converge to the final destination. At high bias (fuel prioritizing), robots show an initial thrust spike, then settle into lower stable thrusts, reducing

fuel consumption. This results in less initial spreading and tighter, more consistent inter-robot formations. Intermediate biases show a mixture of these behaviors.

Reflections

This project helped us to use the concepts learned in class, applying them to real world problems. For example, we can imagine a warzone situation, where the drones must scout dangerous locations, or drop supplies to isolated troops, among other uses.

We were able to approach the problem both theoretically and practically, even though cvxpy revealed itself more difficult than expected, probably due to the lack of familiarity with the Python module. The theoretical questions forced us to not only review the concepts taught in class, but also to deepen our knowledge with research.

In sum, the project was more difficult than expected, but rewarding, in a sense that the results were very satisfying, and corresponded to our previous assumptions.